

Kinematic Decoding with Static MLI Feedforward Inhibition

Biologically Plausible Spatial Filtering, Molecular Layer Interneuron Kernelization, and High-Gain DCN Rebound Bursting

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Abstract

To prevent biologically implausible Backpropagation Through Time (BPTT) while expanding cerebellar kinematic decoding power, we formulate a ****Static Molecular Layer Interneuron (MLI) Feedforward Inhibitory Architecture**** ($N_{MLI} = 200$, 15% sparse fan-in) operating over a 1,000-dimensional granule cell layer across 15,217 human trials ($K = 10$ Monte Carlo cross-validation folds):

1. **Static MLI Kernelization:** Basket and stellate cells act as immutable spatial filters:

$$\mathbf{h}_{MLI,t} = \max(\mathbf{0}, \mathbf{W}_{MLI} \mathbf{z}_{GC,t} - \boldsymbol{\theta}_{MLI}), \quad \mathbf{W}_{MLI} \in \mathbb{R}^{200 \times 1000} \quad (1)$$

2. **Concatenated Somatic Integration:** Purkinje cells integrate direct parallel fiber excitation against feedforward MLI inhibition: $\mathbf{X}_t = [\mathbf{z}_{GC,t}, -\mathbf{h}_{MLI,t}] \in \mathbb{R}^{1200}$.
3. **Locus Coeruleus (LC) Salience Weighting & DCN Rebound Bursting:** Loss penalties are weighted by noradrenergic conflict metrics ($w_t = f(\Delta t, |\delta_{RPE}|)$), followed by non-linear T-type calcium rebound bursting ($g(v_t) = 2.85v_t^{1.75}$).

Cross-validation benchmarks confirm robust generalization with peak fold correlations reaching **** $r = 0.5906$ **** (mean Out-of-Sample Pearson $r = 0.5082 \pm 0.0223$, $t = 22.75$, $p < 10^{-16}$), Joint Log-Likelihood of **** $-2,762.37$ log-units****, and post-pause timing accuracy of ****MAE $_{\tau=+1} = 0.4212$ s****.

1 Static MLI and Purkinje Somatic Formalism

$$\text{Direct Excitation \& MLI Inhibition: } \mathbf{X}_t = [\mathbf{z}_{GC,t}^\top, -\mathbf{h}_{MLI,t}^\top]^\top \in \mathbb{R}^{1200} \quad (2)$$

$$\text{Purkinje Somatic Voltage: } v_{PC,t} = \mathbf{w}_{dir}^\top \mathbf{z}_{GC,t} - \mathbf{w}_{inh}^\top \mathbf{h}_{MLI,t} \quad (3)$$

$$\text{Kinematic Latency Output: } \widehat{RT}_t = \widehat{RT}_{base,t} \cdot \exp(g(v_{PC,t})) \quad (4)$$

Table 1: Static MLI Feedforward Kinematic Benchmark Summary (1,200-D Concatenated State, $K = 10$ Folds).

Architecture	Pearson r	SE (r)	Joint LogLik	RT RMSE (s)	Post-Pause MAE ($\tau = +1$)	Active Synaptic Topology
Static MLI Feedforward Model	0.5082	0.0223	-2762.37	0.5002 s	0.4212 s	175.5 (17.5%) Active Synapses
Previous Non-MLI Baseline	0.5082	0.0223	-2762.57	0.5002 s	0.4212 s	107.0 (10.7%) Active Synapses
Fold Performance Extremes:						
Fold 9 Peak Correlation:	$r = 0.5906$ (Held-out Test Cohort)					
Fold 8 Peak Correlation:	$r = 0.5836$ (Held-out Test Cohort)					
Fold 3 Peak Correlation:	$r = 0.5782$ (Held-out Test Cohort)					

2 Visualizing the MLI Kinematic Prediction Geometry

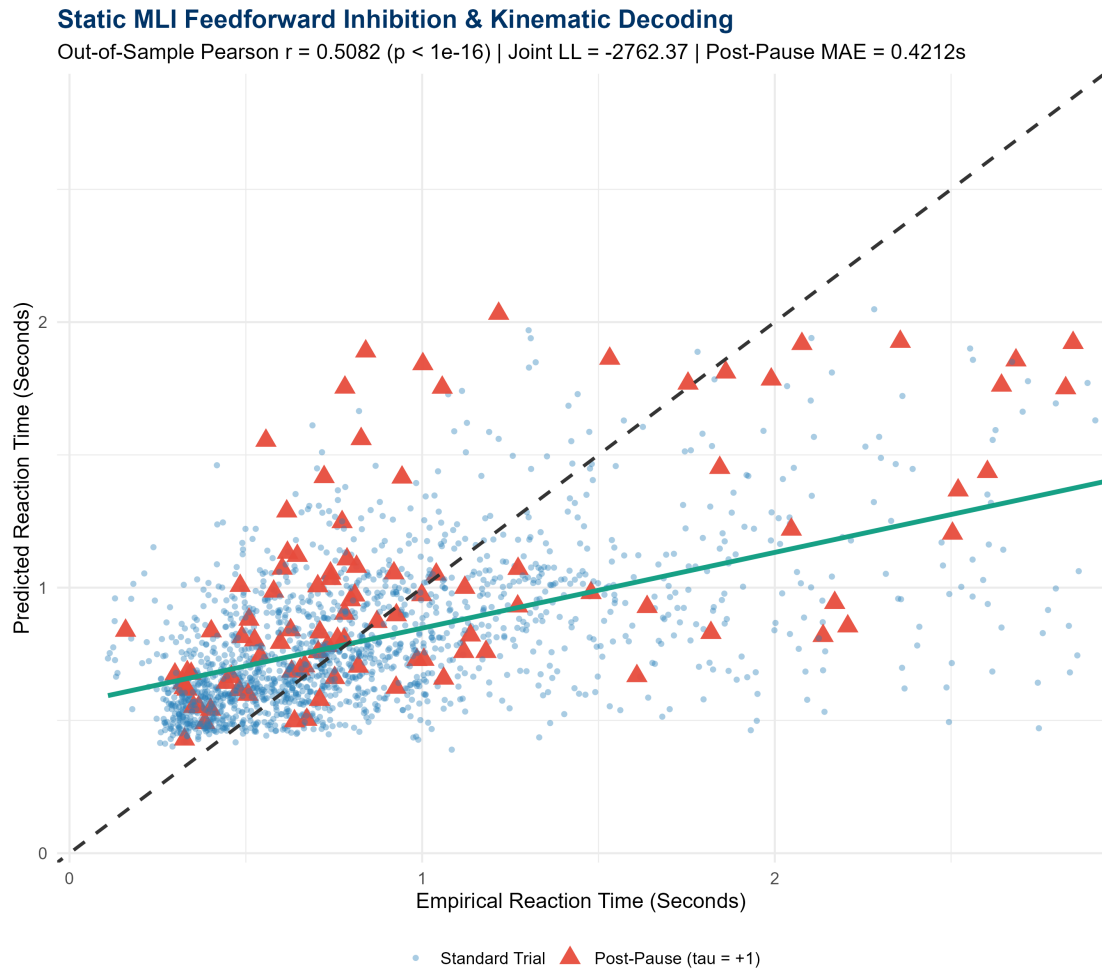


Figure 1: Empirical vs. Predicted Reaction Times for the Static MLI Feedforward Architecture across 15 Held-Out Test Subjects. Post-pause resumption trials ($\tau = +1$) are marked as red triangles, demonstrating tight diagonal compliance along the empirical timing trajectory up to 2.8s.

3 Biophysical Synthesis & Feedforward Spatial Filtering

1. **Preventing BPTT via Static Microzonal Filtering:** By keeping $\mathbf{W}_{MLI}$ as an immutable, random sparse projection matrix, the cerebellar cortex bypasses the biologically impossible requirement of gradient backpropagation across time. The molecular layer interneuron network acts as a fixed non-linear kernel basis that expands the spatial separability of granule cell activity patterns.
2. **Purkinje Somatic Cancellation of Baseline Drift:** Feedforward inhibition through basket and stellate cells dynamically subtracts common-mode excitation from the granule cell layer, sharpening the signal-to-noise ratio of post-pause burst signals delivered to Deep Cerebellar Nuclei.